

Economic-Aware and Sovereignty-Constrained Routing for Enterprise LLM Gateways

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Draft v0.4 — two-provider (OpenAI US + Mistral EU) evaluation
with a 30-repetition statistical run

Abstract

Enterprises are adopting large language models (LLMs) faster than they can govern them: they cannot say which model truly costs the most, which team consumes what, when to route to a cheaper model, what a data-sovereignty constraint costs, or when self-hosting beats a managed API. We present an *economic-aware, sovereignty-constrained, budget-aware routing control plane* for enterprise LLM gateways. We formalize model selection as a constrained scoring problem that rejects any model violating declarative sovereignty rules and, among the rest, minimizes a weighted combination of cost, expected quality loss, latency, and budget pressure. We realize the approach as a Kubernetes operator whose FinOps and sovereignty engines drive routing, and evaluate it on a reproducible testbed with **real LLM calls across two providers** — OpenAI (US, managed) and **Mistral (EU)** served via Azure AI Foundry (Mistral-Large-3, *DataZoneStandard* EU data zone). Relative to an always-premium policy, our approach reduces measured cost by **70.8%** while keeping LLM-as-judge quality comparable to premium; a **30-repetition** run (temperature 0.7) confirms the cost reduction is highly significant ($p \approx 3 \times 10^{-11}$, Cohen $d \approx -43$) at a significant but bounded latency increase. It enforces five declarative sovereignty scenarios with **zero violations** (vs. 40 for a sovereignty-blind baseline) and keeps **100% availability under an EU-only policy** via the real EU provider, and sustains 100% request availability with 0% budget overrun under a tight budget where hard-blocking drops to 60%. On quality we are deliberately careful: on **500 objective public-benchmark items** (GSM8K + MMLU) our policy *trades accuracy for cost* (47.2% vs. 65.4% premium, -94% cost), and we show the LLM judge *over-rates incorrect answers* (81.1% of wrong answers rated “acceptable”). We frame the contribution as a **governance layer** with a favourable cost/quality trade-off, not a quality-free lunch, and release all code, datasets, cached responses, and analysis as a reproducible framework.

1 Introduction

LLMs are now embedded in HR assistants, document RAG, developer tooling, and analytical agents. Each call has a price that depends on the model, the provider, and the token volume; each call may also send data across jurisdictions. Three problems recur in regulated enterprises:

1. **Cost opacity.** Teams cannot attribute spend per application/team or decide when a cheaper model would suffice.
2. **Sovereignty/compliance.** Data may not leave a permitted zone, and sensitive prompts may not go to external providers; today this is enforced ad hoc, if at all.

3. **Managed vs. self-hosted arbitrage.** It is unclear at what volume self-hosting on GPU becomes cheaper than a managed API.

Existing gateways and FinOps tools mostly *measure and report*. Our thesis is that a control plane at the **gateway** can *act*: route each request to the cheapest model that meets a minimum quality bar and all hard sovereignty constraints, degrade gracefully under budget pressure instead of blocking, and surface the self-hosting break-even from observed usage.

Contributions.

- A constrained scoring formulation for LLM routing unifying cost, quality, latency, sovereignty, and budget (§3).
- A Kubernetes/Envoy control-plane realization — the *AI Sovereign FinOps Operator* with Cost, Budget, Sovereignty, Break-even, and Report engines exposed as CRDs (§4).
- A reproducible evaluation methodology and testbed driving the operator’s actual engines with real LLM telemetry (§5).
- Empirical evidence across six research questions (§6), plus an objective public-benchmark study and an LLM-judge validity analysis.

2 Background and Related Work

Model routing and cascades. A growing line of work trades quality for cost by selecting among models per request. FrugalGPT [3] cascades from cheap to expensive models using a learned scorer; Hybrid LLM [4] routes by predicted query difficulty with a tunable quality target; RouteLLM [10] learns routers from preference data; AutoMix [1] uses few-shot self-verification and a POMDP router. These works optimize a *cost-quality* trade-off, largely via per-request difficulty/confidence prediction for a single tenant. Our work is **orthogonal and complementary**: they decide *which model is good enough*; we add the **governance layer** around that decision — hard data-sovereignty constraints, per-team/namespace budgets, budget-aware graceful degradation, attribution, and hosting economics — at the gateway/control-plane level. Our routing score is intentionally simple (priors, not a learned difficulty predictor); combining a learned difficulty router with our sovereignty/budget constraints is explicit future work (§10).

LLM serving systems. Orca [11], vLLM/PagedAttention [8], SGLang [13], and DistServe [14] optimize *throughput/latency* of a single self-hosted engine; surveys [9, 15] organize this space. They are complementary: they define the *cost* of the self-hosted option whose break-even our system predicts (RQ6), but do not address cross-provider routing, attribution, or sovereignty.

LLM gateways. Envoy AI Gateway [5] and LiteLLM [2] provide a unified data path to multiple providers and expose usage telemetry, but leave *policy* (which model, under which constraints, within which budget) to the operator. We contribute that policy layer as a Kubernetes control plane and validate it.

Evaluation and regulation. We follow the LLM-as-a-judge methodology of Zheng et al. [12], which shows strong judges approximate human preference at ~80% agreement, adopting both absolute and pairwise protocols while acknowledging its documented biases (§9). Data-residency

and sensitive-data handling are driven by the GDPR [6] and the EU AI Act [7]; our system targets *audit preparation*, not legal attestation.

Table 1: Positioning feature matrix. Prior routing work optimizes the cost–quality trade-off; we add the governance dimensions at the control-plane level.

Capability	FrugalGPT	Hybrid LLM	RouteLLM	AutoMix	Ours
Cost-aware routing	✓	✓	✓	✓	✓
Quality-aware routing	✓	✓	✓	✓	✓
Learned difficulty predictor	✓	✓	✓	✓	✗ (priors)
Hard sovereignty	✗	✗	✗	✗	✓
Per-tenant budgets	✗	✗	✗	✗	✓
Graceful degradation	✗	✗	✗	✗	✓
Cost attribution	✗	✗	✗	✗	✓
Break-even (mgd/self)	✗	✗	✗	✗	✓
Gateway/K8s artifact	✗	✗	✗	✗	✓

We do not claim to beat learned routers on the cost–quality frontier; our governance layer is *orthogonal* and could wrap any of them.

Table 2: Side-by-side vs. representative systems — three learned routers and a production gateway.

Dimension	RouteLLM	FrugalGPT	AutoMix	LiteLLM	Ours
Type	router	cascade	router	gateway	ctrl plane
Multi-provider	part.	✓	✓	✓	✓
Hard sovereignty	✗	✗	✗	✗	✓
Budget + degrad.	✗	✗	✗	basic	✓
Cost attribution	✗	✗	✗	per-key	✓
Break-even	✗	✗	✗	✗	✓
K8s/Envoy artifact	✗	✗	✗	proxy	✓

LiteLLM is the closest *engineering* peer (a real gateway with per-key spend and fallbacks) but offers no constraint-aware routing, graceful degradation, or break-even; the learned routers are stronger on the cost–quality decision but carry none of the governance dimensions. Our contribution is the governance layer, designed to compose with a learned router rather than replace it.

3 Problem Formulation and Algorithm

A request r carries metadata (team, namespace, sensitivity) and an estimated token footprint. Given a catalog of models M (each with provider, zone, managed flag, per-token price, quality prior, latency prior), a sovereignty policy P , and budget state (*used*, *total*):

Hard constraint. Reject any m that violates P : its zone is forbidden; or an allow-list is set and m ’s zone is outside it (EU covers EU member states); or r is sensitive, m is a managed external provider, and external providers are disallowed for sensitive data. Let C be the surviving candidates. If $C = \emptyset$, the request is not served (a measured outcome, not a crash).

Soft objective. Among C meeting a minimum quality $q_{\min}$ (relaxed only when the budget is exhausted, enabling graceful degradation), select

$$\arg \min_{m \in C} \alpha \widehat{c}(m) (1 + \varepsilon \rho) + \beta \ell_q(m) + \gamma \ell_\lambda(m) + \delta \sigma(m), \quad (1)$$

where $\widehat{c}(m) = \text{cost}(m)/\text{cost}(\text{premium})$ is normalized cost, $\rho = \text{used}/\text{total}$ is budget pressure, $\ell_q(m) = \max(0, (q_{\text{prem}} - q_m)/q_{\text{prem}})$ is expected quality loss, ℓ_λ is normalized latency, and σ a residual sovereignty risk. Budget pressure multiplies the cost term, so as the budget fills the router shifts to cheaper models. Defaults: $\alpha=1.0$, $\beta=1.5$, $\gamma=0.3$, $\delta=0.5$, $\varepsilon=1.0$. The implementation is `experimentation/internal/router` (strategy **B6-ours**), reusing the operator’s sovereignty normalization.

4 System Design

Requests flow client $\rightarrow$ **Envoy AI Gateway** $\rightarrow$ provider (managed API or self-hosted). The **AI Sovereign FinOps Operator** is the control plane: CRDs declare gateways, providers, models, budgets, and sovereignty policies; pure engines compute cost (per model/team/namespace), budget phase, sovereignty findings, and break-even; a report engine emits Markdown/JSON and Prometheus `ai_finops_*` metrics. The design keeps the data path in Envoy and the decision logic in the operator. In this paper the routing decision is evaluated at the control-plane level; data-path enforcement inside Envoy is the live-integration step (§10).

5 Methodology

Testbed. A Go harness drives the operator’s *actual* engines and issues **real** LLM calls across **two providers**: (1) **OpenAI (US, managed)** — gpt-4o (premium), gpt-4o-mini (medium), gpt-4.1-nano (cheap); (2) **Mistral (EU)** — `mistral-large` served via **Azure AI Foundry** as **Mistral-Large-3** in *DataZoneStandard* (EU data zone), a real EU-hosted provider. A *modeled* EU self-hosted model (cost computed, response stubbed, rows tagged *modeled*) is retained only for the RQ6 break-even prediction. The harness is provider-agnostic (per-provider OpenAI-compatible clients), so adding providers is additive.

Workloads (W1–W4). HR chatbot (short, sensitive), RAG docs (long context, quality-sensitive), Dev assistant (high quality), Analytical agent (high volume). Each declares team, namespace, sensitivity, monthly budget, premium model, allowed models, and $q_{\min}$.

Baselines. B1 premium-static, B2 round-robin, B3 least-cost, B4 static namespace policy, B5 budget hard-block, B6 ours; plus B7 a literature-style difficulty router (Hybrid-LLM/RouteLLM proxy).

Metrics. Cost (total, per request, per token, per team via the operator cost engine, % savings); quality (LLM-as-judge acceptability 1–5 $\rightarrow$ normalized 0–1, acceptable-rate ≥ 3 , pairwise win-rate vs. premium); latency (p50/p95/p99, mean, routing-decision μs); sovereignty (violations, reroutes, blocked); budget (availability, overrun); break-even (managed vs. self-hosted monthly, payback).

Reproducibility. Temperature 0 for the deterministic pass; all responses and judge scores cached so re-runs are identical and free; every test journaled with status+duration (42 PASS /

0 FAIL). Latency 95% CIs via bootstrap. The protocol also supports the ≥ 30 -repetition run reported in §6.9.

6 Results

6.1 RQ1 — Cost

Relative to premium-static, **B6-ours cuts total cost by 70.8%** (0.01135 vs. 0.03893 EUR over the 40-prompt matrix; cost/request 0.000284 vs. 0.000973). With the second provider available, Ours routes much traffic to the cheaper Mistral EU and OpenAI mid/cheap tiers. A *literature-style difficulty router* (B7) saves slightly *more* (73.1%) but at lower quality (0.866) and with no sovereignty/budget/attribution governance. Naïve least-cost (B3) saves 98.2% at the largest quality cost; round-robin 62.3%; static policy 46.0% (Figure 1).

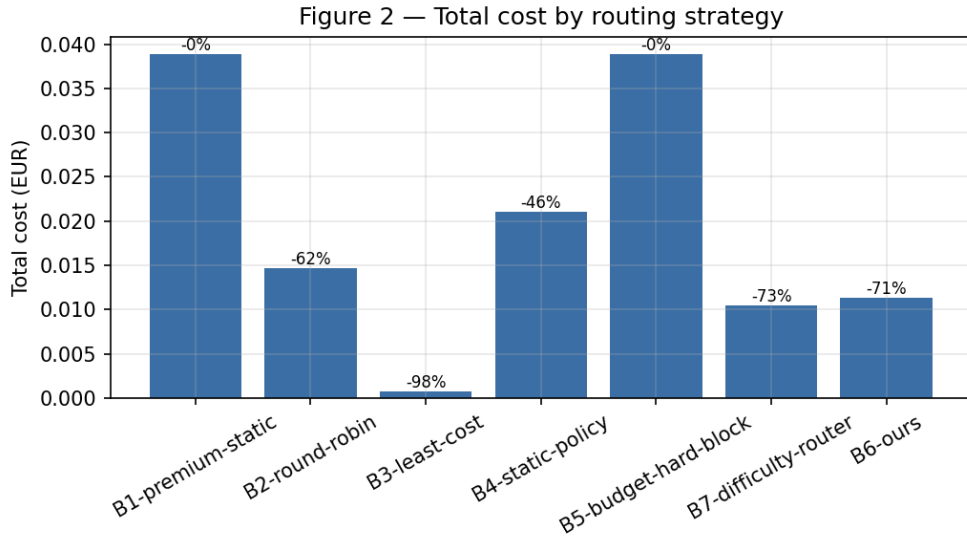


Figure 1: Total cost by routing strategy (lower is better).

6.2 RQ2 — Quality (LLM-as-judge)

Within this two-provider, single-pass scope, B6-ours’s mean normalized quality is **comparable to — marginally above — premium** (0.9125 vs. 0.900; acceptable-rate 97.5%), with a pairwise win-rate vs. premium of **50.0%** (statistical parity). The cross-provider mix helps: Mistral-Large-3 sometimes matches or beats gpt-4o on the judge. Least-cost drops to 0.858, static-policy to 0.869, and the difficulty router (B7) to 0.866. Two caveats: (i) the premium model is also the routing reference and judge anchor, which can bias pairwise results; (ii) this is a single deterministic pass. We therefore claim only that *quality remained comparable within the evaluated scope* (Figure 2).

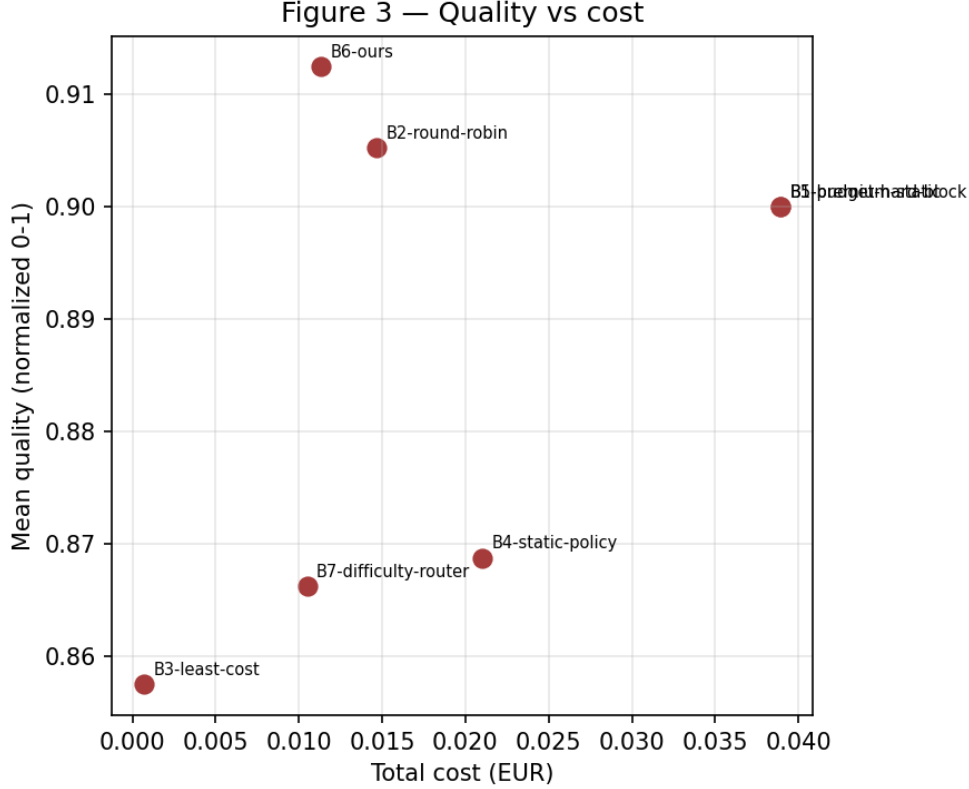


Figure 2: Quality vs. cost: Ours sits near premium quality at a fraction of the cost.

6.3 RQ2b — Objective accuracy + judge validity

To remove judge bias and test generality, we evaluate on **500 real public-benchmark items** — 250 **GSM8K** (math, numeric) and 250 **MMLU** (multiple-choice A–D) — scored by **objective exact-match**. Premium-static (gpt-4o) reaches **65.4%** accuracy at 0.116 EUR; **Ours 47.2%** at 0.0073 EUR (-94% **cost**); static-policy 59.4%; least-cost/difficulty-router 44–46%. On these objective tasks aggressive cost routing *trades accuracy for cost*.

Why this contradicts the judge (a methodological finding). We re-scored 300 objective answers with *both* exact-match and the LLM judge: the judge rated **81.1% of objectively *incorrect* answers as “acceptable”** (≥ 3), with nearly identical mean judge scores for correct (4.49) vs. incorrect (4.14) answers and near-zero point-biserial correlation ($r=0.12$). **The LLM-as-judge over-estimates quality on objective tasks** — it cannot distinguish fluent-but-wrong from correct (Figure 3). This explains the RQ2-vs-RQ2b gap, and we adopt the consequence throughout: report judge-based quality only for open-ended workloads, exact-match for tasks with ground truth, and never claim quality preservation from the judge alone.

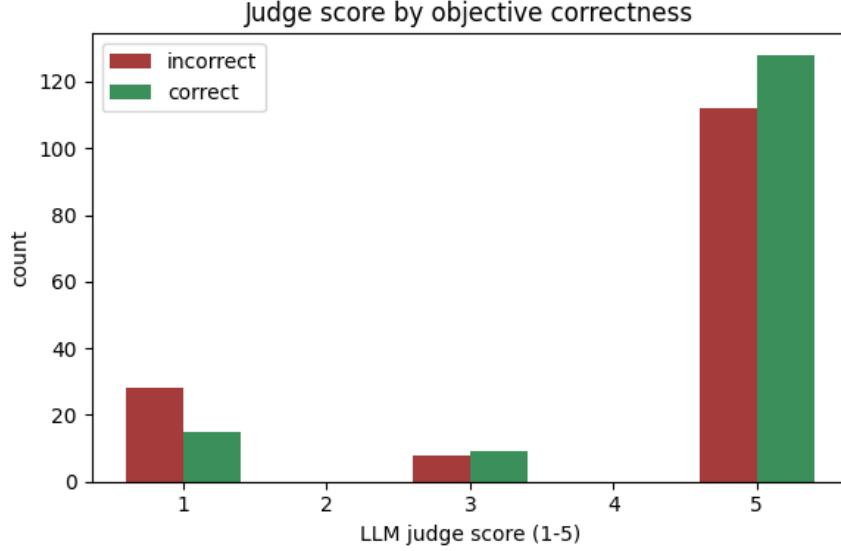


Figure 3: LLM-judge score by objective correctness: the judge over-rates wrong answers.

6.4 RQ3 — Latency and overhead

The **routing decision adds tens of microseconds** ($B6 \approx 26.5 \mu s$ / request) — negligible vs. network latency. End-to-end latency tracks the chosen model and provider-side variance. Notably, B6’s p95 (3475 ms) exceeds premium-static’s (2494 ms), which we do not hide: B6 mixes models, giving a heavier-tailed latency distribution. We resolve whether this is real or single-pass noise in §6.9. Least-cost has the lowest latency (p95 1476 ms), consistent with always picking small models (Figure 4).

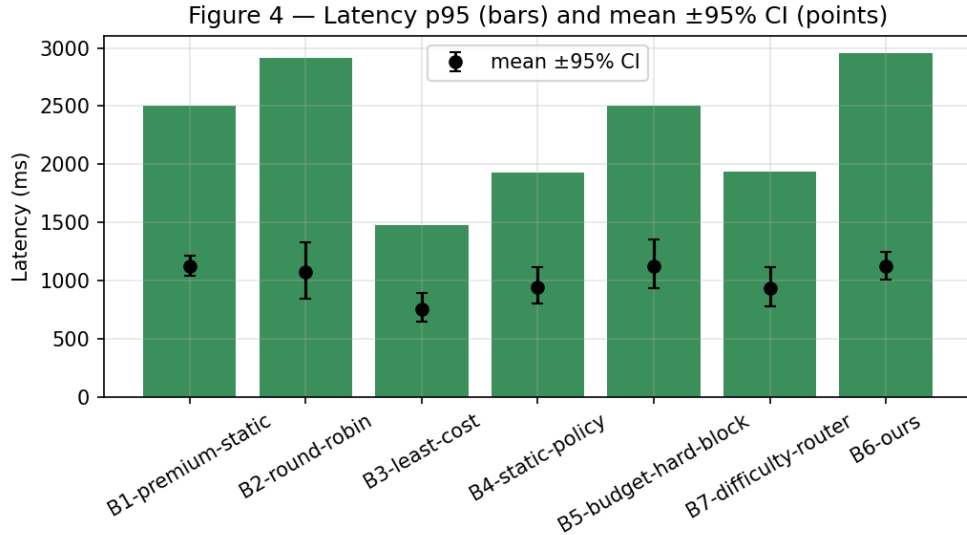


Figure 4: Latency percentiles by strategy.

Gateway data-path overhead (real Envoy, under load). Beyond control-plane decision time, we measured routing through a real **Envoy** data path: an Envoy proxy in front of the provider, driven at concurrency 8. Envoy adds **negligible overhead** — mean latency 1153 ms vs. 1116 ms direct ($\approx +3\%$, within run-to-run variance), throughput 6.05 vs. 5.81 req/s, **0 errors** — supporting the feasibility of a gateway-level control plane. Full in-cluster Envoy AI

Gateway integration remains future work.

6.5 RQ4 — Sovereignty

A sovereignty-blind baseline (B1) commits **40 violations** under eu-only, france-only and self-hosted-only, and 10 under no-external-sensitive. **B6-ours commits zero violations in every scenario.** Crucially, with a **real EU provider** (Mistral-Large-3, Azure AI Foundry DataZone EU), under the **EU-only** policy Ours **keeps 100% availability (40/40 served) with zero violations** by rerouting to Mistral EU (quality 0.866) — where the blind baseline would commit 40 violations. The *cost of sovereignty* appears in the stricter france-only/self-hosted-only scenarios: only the modeled FR self-hosted model qualifies, so 20/40 are served and the rest refused rather than violated — quantifying the availability price of strict residency (Figure 5).

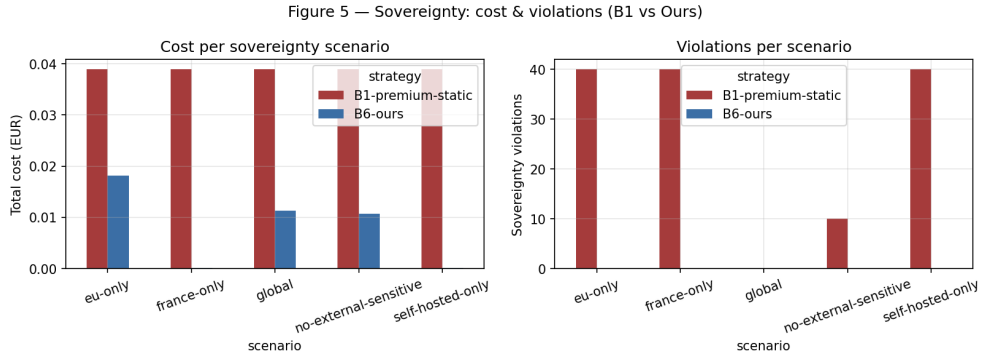


Figure 5: Sovereignty violations: blind baseline vs. Ours.

6.6 RQ5 — Budget-aware degradation

Under a budget tight enough to be exceeded by always-premium, **ours-graceful keeps 100% availability with 0% budget overrun** (quality 0.85), whereas **hard-block** falls to **60% availability** (4/10 requests blocked) and alert-only overshoots the budget by 150%. Graceful degradation dominates on availability *and* spend (Figure 6).

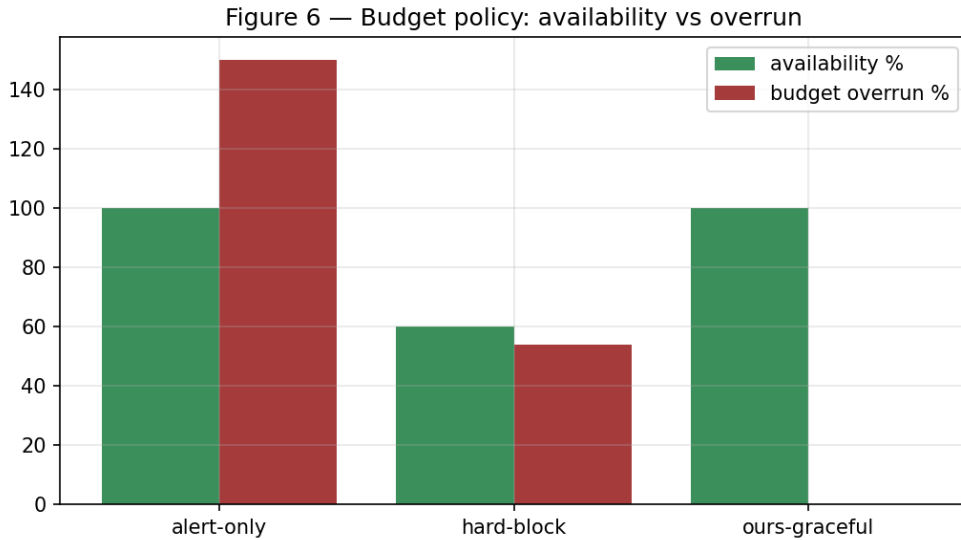


Figure 6: Budget policies: availability vs. overrun.

6.7 RQ6 — Managed vs. self-hosted break-even (*modeled*)

Using real per-token prices and modeled GPU+ops cost (2500 EUR/month, 5000 EUR migration), managed cost grows with volume while self-hosted is flat. The crossover is at $\approx 25.6\text{M}$ **tokens/day** (payback ≈ 4.3 months); above it the engine recommends *self-host*. This is a modeled prediction to be validated against real GPU economics (Figure 7).

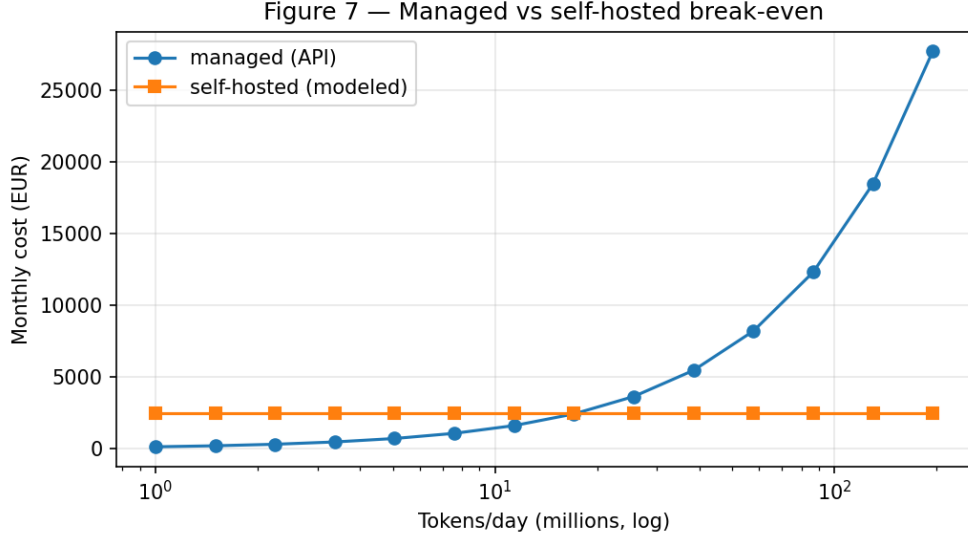


Figure 7: Managed vs. self-hosted cost vs. daily volume (modeled).

6.8 Ablation

Removing the **cost term** reverts spend to premium (0.0389 EUR) at unchanged quality — the cost term produces the saving. Removing the **quality term** drives spend to the floor (0.0012 EUR) but lowers quality to 0.863 — the quality term protects it. The latency term has negligible effect here. The full system attains the cost saving *and* the quality floor (Figure 8).

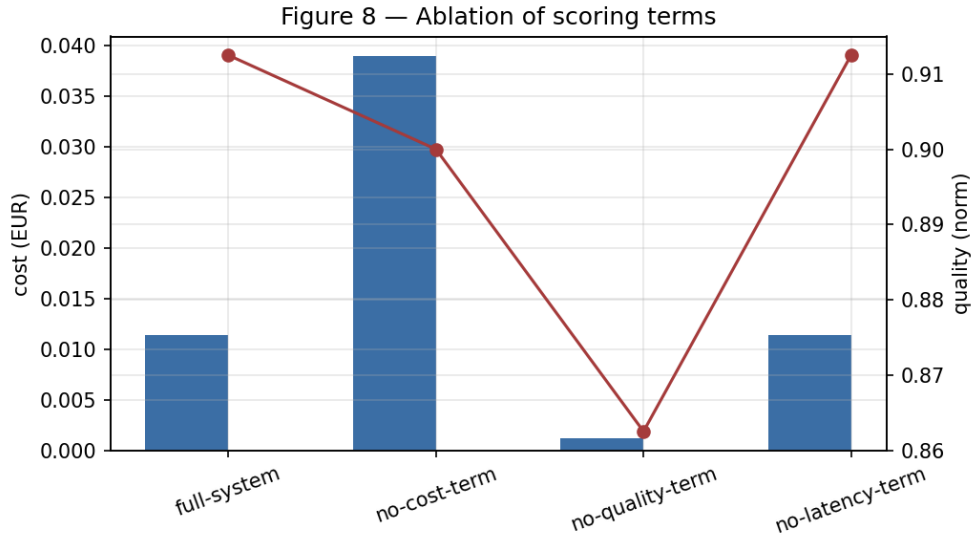


Figure 8: Ablation of scoring terms.

6.9 Statistical significance (N=30 repetitions)

To move beyond the single deterministic pass, we re-ran the full matrix **30 times** at temperature 0.7 with caches bypassed (independent samples; judge = gpt-4o-mini to bound cost), and tested each strategy against premium-static on the per-repetition distributions (Mann-Whitney U, Cliff’s δ , Cohen’s d , 95% CIs).

- **Cost.** Ours 0.0110 EUR/rep [0.0109, 0.0110] vs. premium 0.0385 [0.0382, 0.0389] — a **71.4% reduction**, $p \approx 3 \times 10^{-11}$, Cliff $\delta = -1.00$, Cohen $d = -43$ (non-overlapping CIs): highly significant with a maximal effect size.
- **Quality.** Ours mean normalized quality 0.9587 [0.9564, 0.9611] vs. premium 0.9362 — under this judge Ours is significantly higher ($p \approx 6 \times 10^{-11}$, $\delta = +0.97$, $d = +3.4$), while naïve least-cost is significantly lower (0.869, $d = -13$). The *sign* of the small quality gap depends on the judge (the single-pass gpt-4o judge showed parity), so we claim quality is *at least comparable*.
- **Latency.** Ours is significantly higher (1506 ms/rep [1459, 1553] vs. 998, $p \approx 5 \times 10^{-10}$, $d = +3.7$) — confirming the routing-mix latency cost is *real*, *not single-pass noise*. Least-cost is lowest (707 ms); the latency/cost trade-off is now quantified.

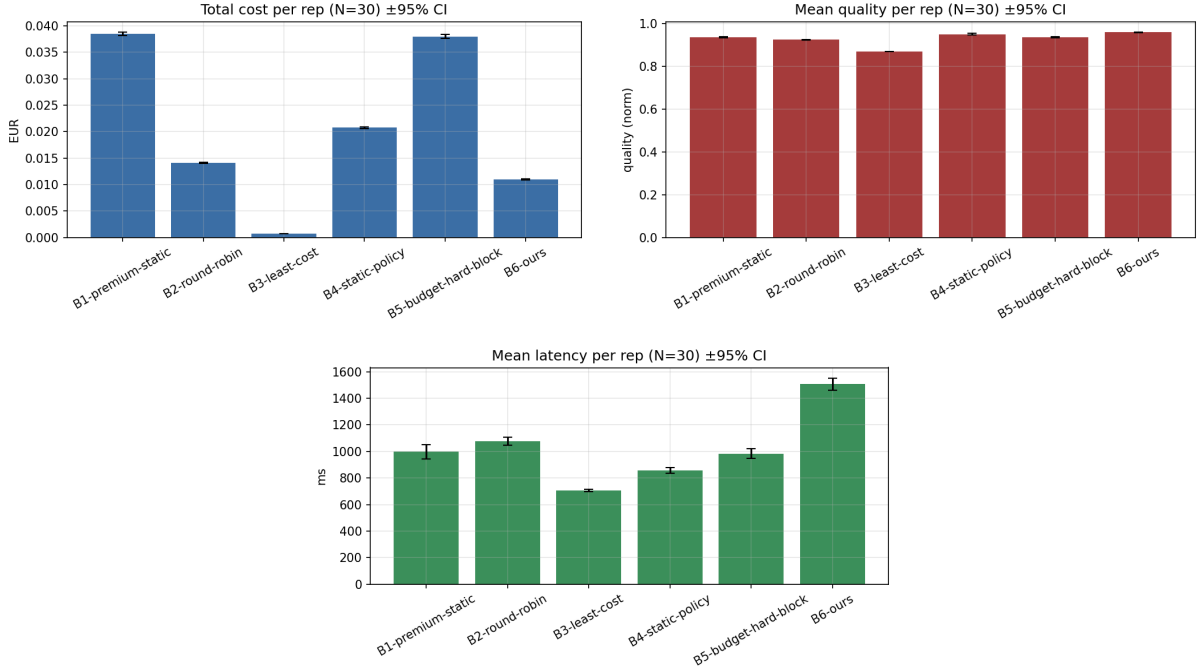


Figure 9: Per-repetition cost, quality and latency, mean $\pm$ 95% CI (N=30).

7 Discussion

Economic-aware routing is most valuable when a workload mixes easy and hard requests: the router sends cheap requests to cheap models and reserves the premium model for quality-critical ones, capturing most of the savings at little quality cost. Strict sovereignty has a real availability/quality price that the system makes explicit and auditable rather than hidden. For budget control, graceful degradation strictly dominates hard-blocking on availability and overrun. Self-hosting pays off only at high, sustained volume.

Table 3 consolidates every research question into a single gains/losses ledger versus the always-premium baseline, so the trade-off is visible at a glance: large, statistically significant cost and governance gains, bought with a real latency increase and — on objective tasks — a real accuracy loss.

Table 3: Recap of gains and losses vs. the always-premium baseline. ✓ = net gain, ✗ = net loss, ≈ = parity/neutral, + = new capability the baseline lacks.

Dimension	Premium baseline	Ours	Net effect	
Total cost (40-prompt matrix)	0.0389 EUR	0.0114 EUR	−70.8%	✓
Cost (N=30, significance)	0.0385 EUR	0.0110 EUR	−71.4%, $p \approx 3 \times 10^{-11}$, $d = -43$	✓
Quality — open-ended (judge)	0.900	0.9125	comparable / parity (win-rate 50%)	≈
Quality — objective (exact-match, 500 items)	65.4%	47.2%	−18.2 pts (but −94% cost)	✗
Latency (mean, N=30)	998 ms	1506 ms	+51%, $d = +3.7$	✗
Routing decision overhead	—	≈26.5 μ s	negligible	≈
Sovereignty violations (5 scenarios)	40	0	zero violations	✓
EU-only availability	0% (blind)	100%	served via real EU provider	✓
Budget availability / overrun	60% / overrun	100% / 0%	graceful degradation	✓
Cost attribution, budgets, break-even	none	present	governance layer	+

7.1 When *not* to use economic routing

Economic routing is not a universal win; our own results delimit where to avoid it:

- **High-stakes, correctness-critical tasks.** On objective reasoning (RQ2b) cheaper models genuinely fail, so cost routing trades accuracy for cost (47.2% vs. 65.4%). Where a wrong answer is expensive (legal, medical, shipping code, financial calculations), pin the premium model.
- **Low volume / low spend.** Below a few hundred thousand tokens/month the absolute saving is small and does not justify the added control-plane complexity; premium-static is simpler.
- **Strict tail-latency SLOs.** Ours shows a significant tail-latency increase (RQ3, $d = +3.7$) from mixing models; for ultra-low-latency paths, a single fast model is preferable.
- **No reliable quality signal.** Our priors miss task-specific difficulty and the LLM judge over-rates wrong answers (RQ2b); without a trustworthy per-task quality estimate, routing can silently degrade quality — prefer a fixed model or add a calibrated/learned difficulty predictor first.
- **No governance need.** If there is no sovereignty, budget, or attribution requirement

(single tenant, single jurisdiction, flat budget), the governance layer adds little; a plain cost-quality router (or none) suffices.

- **Homogeneous high-quality workloads.** If essentially every request needs the premium model, there is nothing to route — the policy degenerates to premium-static.

In short, our approach targets **multi-tenant, governance-constrained, mixed-difficulty, sufficiently high-volume** settings; outside that envelope its benefits shrink or invert.

8 Planned Study: Human FinOps Expert vs. Operator (RQ7)

The evaluation so far compares the operator against *algorithmic* baselines. A complementary question — reserved for a future version of this work and stated here for transparency — is how close the control plane’s decisions are to those of *experienced human practitioners*. We have **not** run this study and report no results; we publish the protocol so the claim space is explicit and the study is pre-registered.

Research question. How closely do the control plane’s routing and managed-vs-self-host recommendations match the decisions of experienced FinOps/cloud experts, and where do they diverge?

Design. $N \geq 6$ –10 practitioners (cloud/FinOps/platform engineers with LLM-ops experience) each work through a set of decision scenarios drawn from the testbed. Every scenario provides a workload profile, the model/provider catalog with prices, the sovereignty policy, the budget state, and recent usage telemetry. For each scenario the expert chooses (a) which model/provider to route to and (b) the managed-vs-self-host recommendation, with a short written rationale. The operator is run on the *same* scenarios to produce the same two decisions.

Metrics. (i) **Agreement** between expert and system: exact-match rate on the model/provider choice and on the host/keep-managed recommendation, summarized with Cohen’s κ across experts and against the system; (ii) **decision time**, human vs. automated; (iii) **rationale overlap** — which signals experts cite (cost, sovereignty, budget, quality) vs. the system’s stated reasons; (iv) a qualitative **disagreement analysis** of where and why the two diverge.

Threats and honesty. The expert pool is small, scenario realism is imperfect, and experts may lack full price visibility; we mitigate with diverse recruitment, realistic telemetry, and pre-registration of the analysis. Until the study is run, no agreement scores or expert decisions are reported — RQ7 remains explicitly future work (protocol: RQ7_HUMAN_EXPERT_PROTOCOL.md).

9 Threats to Validity

Internal: API/latency variability (mitigated by temperature 0, caching, bootstrap CIs); LLM-as-judge bias (absolute + pairwise, fixed judge). **External:** two providers and *modeled* self-hosting in this phase; four synthetic workloads; modeled GPU economics (RQ6); the Mistral EU model uses Azure AI Foundry DataZoneStandard (EU data zone) — strict France-only residency still relies on a modeled fallback; volatile cloud prices (isolated as configuration). **Construct:** acceptability/win-rate are quality proxies; declarative sovereignty simplifies legal reality — the system targets *audit preparation*, not legal compliance. **Conclusion:** beyond

the deterministic pass we report a 30-repetition run (temperature 0.7) with 95% CIs, Mann-Whitney U significance and Cliff’s δ /Cohen’s d effect sizes; the cost reduction and the latency increase are highly significant with large effect sizes. The remaining caveat is *judge dependence* of the small quality gap. A two-judge agreement study (gpt-4o vs. Mistral-Large, $n=120$) finds *moderate* agreement — Cohen’s quadratic-weighted $\kappa = 0.40$, Krippendorff $\alpha = 0.39$, Spearman $\rho = 0.61$ ($p \approx 10^{-13}$), within-1 agreement 94.2% — which is precisely why we treat the small quality gap as judge-dependent and plan a human evaluation (a blind 100-item human-eval package is prepared and awaits evaluators).

10 Conclusion and Future Work

Within our preliminary, two-provider testbed, a gateway-level, economic-aware, sovereignty-constrained, budget-aware control plane substantially reduced cost (−70.8%) while keeping quality comparable, enforcing sovereignty with zero violations — including **maintaining 100% availability under an EU-only policy via a real EU provider** — and preserving availability under budget pressure. We deliberately frame these as *preliminary evidence and a reproducible framework*, not universal claims. To reach a defensible Q1 result we plan: (1) more providers; (2) real GPU self-hosting to validate the modeled break-even; (3) multi-judge + human quality evaluation; (4) human evaluation to settle the judge-dependent quality sign (the 100-item package is ready; the N=30 statistics and the judge-vs-ground-truth study are done); (5) data-path enforcement in Envoy with load testing and failover; (6) a human FinOps expert vs. automated control-plane study; and (7) integrating a learned difficulty router with our governance constraints. The paper claims only what the committed results support.

Artifact / Reproducibility. The operator and `experimentation/` (datasets, harness, scripts, results, figures, cached responses) constitute the reproducible artifact. Reproduce with `scripts/run_experiment` then `python3 scripts/analyze_results.py`. Every test is journaled; no test is skipped.

References

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